Yuanshuo Du

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Summary

Full-stack Software Engineer with a security-first mindset and penetration-testing background. Deployed CNN models achieving 95% accuracy for satellite-based parking detection, reducing manual survey labor by 60%. Built Kubernetes microservices with automated CI/CD, cutting release cycles from 2 days to 15 minutes. Secured 20+ client systems across finance, healthcare, and technology sectors; aligned technical controls with ISO 27001 requirements across 5+ certification projects.

Skills

Languages: Python JavaScript TypeScript SQL HTML/CSS
Frameworks: React Node.js Express Redux FastAPI Django
Cloud/DevOps: Azure Kubernetes Docker Terraform GitHub Actions CI/CD
Databases: PostgreSQL Redis MongoDB
Security: Burp Suite Metasploit Wireshark Nmap OWASP ZAP ISO 27001 Audit Penetration Testing Secure Systems Design

Experience

Information Cybersecurity Specialist — InsightSec, Shanghai
Penetration Testing & Compliance Consulting — Apr 2021 – Dec 2022

- Led 20+ penetration testing engagements (OWASP PTES / NIST frameworks), remediated 20+ critical vulnerabilities — reduced client risk exposure by 70% across finance, healthcare, and technology sectors
- Coordinated cross-functional Agile teams across 5+ ISO 27001 certification projects, aligning technical controls with business requirements — all delivered on time, within budget, zero major non-conformities
- Streamlined audit documentation workflows, cutting compliance overhead by 40%

Web Engineer — Sheffield African Caribbean Community Association
Full-Stack Web Development — Sep 2019 – Apr 2020

- Designed and built a responsive, mobile-first website (WordPress + Figma) from scratch — organic search visibility grew 40% in 6 months; average session duration rose from 1.2 min to 2.8 min
- Refactored legacy front-end components, reducing maintenance overhead by 30% and improving load times across all device categories

Technical Projects

Smart Parking Space Detection (Magpie) — Sep – Dec 2024 — *Next.js, Node.js, PostgreSQL, CNN, Kubernetes, CI/CD*

- Trained CNN on 10,000+ satellite images (ResNet50 transfer learning) — 95% classification accuracy, reducing manual survey labor by 60%; deployed containerised microservices on Kubernetes with fully automated GitHub Actions CI/CD, cutting release cycles from 2 days to 15 minutes

AI-Powered Q&A Platform — Mar – Sep 2024 — *H5, UniApp, RAG, PostgreSQL, GitBook*

- Architected RAG-powered retrieval pipeline (semantic embeddings + BM25 hybrid search) in an H5/UniApp app — reduced user search time by 45% vs. keyword-only baseline; led cross-functional team of 3 through full SDLC with 4.7/5 stakeholder satisfaction; achieved 92% bug-free production rate across 2 major releases

Travel Planner (MERN Full Stack) — Sep – Dec 2023 — *React, MaterialUI, Redux, Node.js, Express, MongoDB*

- Built production MERN app with JWT + Google OAuth 2.0, WebSocket real-time commenting, and optimised MongoDB indexing — served 500+ users; 4.6/5 average session rating; page load reduced from 3.2s to 0.8s

Education

MSc Computer Science — Technological University Dublin — Dec 2024
Software Design • Advanced Databases • Systems Architectures • Secure Systems Dev • Deep Learning • 2:1 (Merit)

MSc Management (International Business) — University of Sheffield — Apr 2021
CMI Level 7 Award in Strategic Management and Leadership

BA English and American Studies — University of Pannonia (Erasmus+) — Jun 2018
GPA: 4.11/5

Certifications & Publication

ISO 27001 Internal Auditor (BSI) ▪ CMI Level 7 Strategic Management ▪ PRINCE2 Practitioner

Du, Y. et al. (2025). The Development of a GIS for Public Amenities in Urban Areas. *EduLearn25*.